SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA37- SOFTWARE TESTING

EXPERIMENT 5

Case Study: E-Commerce Application – Negative Test Scenarios and Test Cases

AIM:

To design and execute negative test scenarios and test cases for an E-Commerce Application to verify its behavior under invalid inputs, unavailable products, incorrect quantities, invalid delivery details and payment failures, and to identify and report software defects.

Problem Description

An online shopping application allows customers to register, log in, search for products, add products to a shopping cart, and place orders. Customers must provide valid details during registration and login. The application should validate product availability, quantity, delivery address, and payment details before confirming an order. The testing team is responsible for verifying whether the application works correctly under valid and invalid conditions. They must prepare positive and negative test cases and document any defects identified during testing.

Functional Requirements

1. Users must register with a valid name, email address, password, and mobile number.

2. Users must log in using valid credentials.

3. Products must be added to the cart only when they are available.

4. The requested quantity must be greater than zero and must not exceed available stock.

5. The application must accept valid delivery details.

6. An order must be confirmed only after successful payment.

7. The application must display appropriate error messages for invalid inputs.

3. Negative Test Cases

Negative testing verifies how the application behaves when invalid, incomplete, or unexpected inputs are provided.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
NT01	Register without an email	Leave email blank	Validation message displayed	Pass
NT02	Register with invalid email	user@invalid	Email format error displayed	Pass
NT03	Login with incorrect password	Correct email, wrong password	Login denied with error message	Pass
NT04	Add an unavailable product	Select an out-of-stock product	Product cannot be added or ordered	Pass

NT05	Enter zero quantity	Set quantity to 0	Quantity rejected	Pass
NT06	Enter quantity exceeding stock	Request 10 units when only 5 are available	Quantity rejected with appropriate message	Fail
NT07	Enter an invalid PIN code	Enter letters in PIN field	Invalid PIN rejected	Pass
NT08	Payment failure	Simulate declined payment	Order not confirmed; failure message displayed	Pass
NT09	Submit empty registration form	Leave all mandatory fields blank	Required-field messages displayed	Pass
NT10	Access order history without login	Open order history while logged out	User redirected to login	Pass

4. Detailed Execution of a Failed Test Case

Test Case NT06
Failed

Test scenario: Enter quantity exceeding stock

Precondition

Wireless Mouse has only 5 units in stock.

Test steps

1. Log in.
2. Search for Wireless Mouse.
3. Open product page.
4. Enter quantity 10.
5. Click Add to Cart.

Expected result

Application should reject the quantity and display: “Only 5 units are available.”

Actual result

Application accepts all 10 units and adds them to the cart.

Conclusion

The test case fails because the application does not enforce the stock quantity limit.

5. Sample Bug Report

A bug report is a document that records a defect discovered during software testing. It provides developers with sufficient information to reproduce, investigate, and fix the issue.

Field	Details
Bug ID	BR-001
Linked Test Case	NT06
Severity	High
Priority	High
Bug Title	Enter quantity exceeding stock
Environment	Application: E-Commerce Application; Browser: Google Chrome; Operating System: Windows 10; Test Module: Relevant application module

Preconditions	Wireless Mouse has only 5 units in stock.
Steps to Reproduce	1. Log in. 2. Search for Wireless Mouse. 3. Open product page. 4. Enter quantity 10. 5. Click Add to Cart.
Expected Result	Application should reject the quantity and display: “Only 5 units are available.”
Actual Result	Application accepts all 10 units and adds them to the cart.
Impact	The defect may cause incorrect system behavior, data inconsistency, transaction errors, or poor user experience.
Suggested Fix	Apply validation on the client and server side and repeat the validation at the final submission/transaction stage.
Status	Open — awaiting developer investigation and correction.

6. Bug Life Cycle

New — Tester reports the defect
Assigned — Developer receives the defect
In Progress — Developer investigates and fixes it
Fixed — Developer submits the correction
Retesting — Tester verifies the correction
Closed — Tester confirms the defect is resolved
If the defect still exists after the fix, the tester reopens the bug and sends it back for further correction.

7. Difference Between Positive and Negative Testing

Positive Testing	**Negative Testing**
Uses valid inputs.	Uses invalid or unexpected inputs.
Checks whether expected operations succeed.	Checks whether errors are handled correctly.
Example: valid login or valid transaction.	Example: wrong password or invalid transaction.
Expected result is successful execution.	Expected result is safe rejection or appropriate error handling.